

A closure study to validate a novel isotope-aware Lagrangian cloud-microphysics model against airborne measurements

1 Scientific goal of the project

Stable water isotopologues are tracers of atmospheric water cycle processes, as isotopic fractionation during phase changes and transport preserves information on cloud thermodynamic and microphysical history.

The project aims to reconstruct cloud isotopic processes using particle-resolved numerical simulations constrained by below-cloud vapour isotopic measurements, which serve as boundary and validation data.

A key question is whether isotopic observations can constrain cloud microphysics sufficiently to achieve *isotopic closure* in a prognostic particle-based framework. It remains unknown whether closure is achievable with current observations and modelling tools, and what minimal model complexity is required.

Supersaturation variability plays a central role in condensation and evaporation at the particle level, strongly influencing both microphysics and isotopic partitioning (Fahandezh Sadi et al. 2026).

The project addresses:

- Q1** Can isotopic closure be achieved using available observations in a particle-based model?
- Q2** Which microphysical processes imprint detectable isotopic signatures?
- Q3** How strongly do isotopic measurements constrain supersaturation, mixing, and phase partitioning?
- Q4** What is the minimal framework required for isotopic closure with predictive skill?

2 Significance of the project

Cloud microphysical processes are a major source of uncertainty in weather and climate projections (Intergovernmental Panel on Climate Change 2021). Key processes such as condensation, evaporation, entrainment, and phase transitions strongly affect cloud evolution and radiative properties, yet remain poorly constrained by observations (Grabowski and Wang 2013). Stable water isotopologues provide tracers of cloud evolution because isotopic composition reflects thermodynamic and microphysical history (Bolot et al. 2013). Although isotopes are widely used in atmospheric models (Galewsky et al. 2016), most approaches rely on Eulerian bulk parameterisations that cannot resolve particle-scale processes controlling isotopic signals (Joussaume et al. 1984; Yoshimura et al. 2008; Werner et al. 2011; Galewsky et al. 2016).

In contrast, particle-based models explicitly represent droplets and ice particles and their microphysics (Shima et al. 2009), offering a natural framework for isotope-resolved cloud modelling. A key open question is whether isotopic observations can constrain cloud microphysics sufficiently to achieve isotopic closure, analogous to CCN closure studies (Knopf et al. 2021). Despite progress in both observations and modelling, no systematic framework exists to evaluate isotopic closure.

This project addresses this gap by combining particle-based simulations with in-cloud and laboratory isotopic observations, linking isotope measurements directly to cloud microphysical processes.

3 Research Concept and Work Plan

The project is structured into three work packages: theoretical development (A1), numerical implementation (A2), and observational validation (A3). Iterative feedback between them ensures consistency between model formulation and interpretation of isotopic signals.

The work builds on preliminary NCN results in isotope-enabled particle-based cloud modelling within the Super-Droplet Method framework, providing a proof-of-concept for prognostic isotopic tracking.

A1 Theory: isotopic cloud microphysics

This work develops a unified theoretical description of isotopic fractionation in liquid–vapour systems, with extension to ice-phase processes. The goal is to derive governing equations for isotope-specific mass transfer in clouds, including equilibrium and kinetic effects. The analysis includes: (i) isotopic mass balance equations for vapour, liquid, and ice; (ii) limiting regimes such as equilibrium fractionation and Rayleigh distillation; and (iii) dimensionless control parameters, including the Bolin number introduced in preliminary work.

A key objective is to identify the minimal set of processes governing isotopic signatures in clouds, including condensation, evaporation, deposition, sublimation, and freezing, formulated for particle-based models. The outcome is a closed theoretical framework suitable for numerical implementation.

The framework also includes analysis of model errors, parameter sensitivity, and uncertainty propagation. This covers both local sensitivity of governing equations and ensemble-based global sensitivity studies, quantifying the impact of microphysical parameters, initial conditions, and boundary conditions. A distinction is made between structural and observational uncertainty.

Preliminary results

A dimensionless description of isotopic evolution in particle-based cloud models was introduced via the Bolin number, extending the work of Bolin (1958). It quantifies the relative importance of diffusive versus phase-change-driven isotopic exchange: $\mathbf{Bo} = \frac{\tau_{\text{heavy}}}{\tau_{\text{light}}}$, where $\tau = \left(\frac{1}{m} \frac{dm}{dt}\right)^{-1}$. This formulation allows inference of heavy isotopologue evolution from the corresponding light species and captures equilibrium fractionation (α_{eq}), diffusion, and environmental controls within a unified framework.

The Bolin number reproduces equilibrium fractionation curves for vapour and liquid phases from Gedzelman and Arnold (1994) (Fig. 1) and has been accepted for presentation at the AMS Madison Summit 2026.

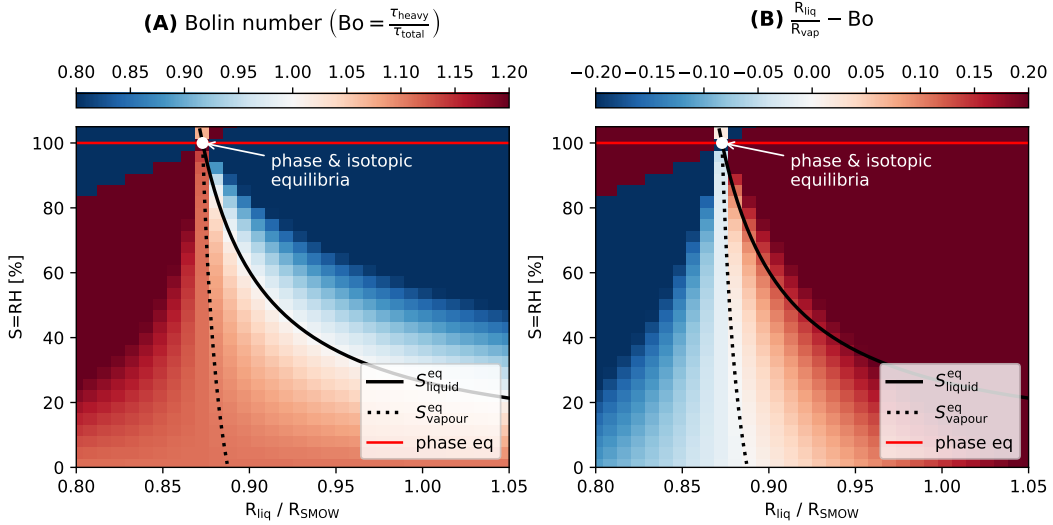


Figure 1: Isotopic steady-state conditions in liquid droplets and vapour. Panel A shows $dR_{\text{liq}}/dt = 0$ corresponding to $\text{Bo} = 1$. Panel B shows $dR_{\text{vap}}/dt = 0$ corresponding to $R_{\text{liq}}/R_{\text{vap}} - \text{Bo} = 0$. Black lines indicate equilibrium curves from Gedzelman and Arnold (1994).

A2 Numerics: particle-based implementation and simulation framework

State of the art

The Super-Droplet Method (SDM) represents cloud and aerosol populations as Lagrangian computational particles with multiplicities (Shima et al. 2009). PySDM is an open-source SDM implementation enabling

simulation of condensation, evaporation, collision–coalescence, and ice-phase processes in various configurations (Bartman et al. 2022; de Jong et al. 2023). Its structure allows straightforward extension with additional prognostic variables such as isotopic composition.

Preliminary results

The PI has implemented isotopic evolution of vapour and droplets for a single heavy isotopologue in PySDM, enabling prognostic isotopic composition of individual super-droplets and explicit representation of fractionation during microphysical processes.

Planned work

The model will be extended to a fully mixed-phase, multi-isotopologue framework. Currently, isotope tracking exists only for vapour and liquid phases; ice-phase representation will be added. The extended formulation will enable consistent treatment of fractionation during phase changes, diffusion, and collisions in a particle-resolved framework.

The implementation will be verified using conservation diagnostics and limiting analytical cases. It will then be explored in idealised box, parcel, and column simulations to quantify sensitivity to supersaturation, temperature, droplet size distribution, and ice fraction. The outcome is a fully mixed-phase isotope-enabled PySDM model for cloud process studies.

A3 Data: observational and experimental validation

Model evaluation will use isotopic observations from aircraft campaigns, including CAESAR, and laboratory experiments conducted with NSF NCAR (Elise Rosky). These datasets provide constraints across vapour, cloud water, ice, and precipitation phases.

The laboratory component is currently preliminary and will evolve alongside model development, enabling iterative feedback between modelling and experiment design. This ensures consistency between observational capabilities and modelling needs while supporting interdisciplinary collaboration.

The analysis focuses on: (i) model–data agreement, (ii) sensitivity of isotopic signals to microphysical processes, and (iii) the ability of isotopic measurements to constrain cloud evolution and assess closure. The outcome is an empirical assessment of isotopic closure and the information content of isotopic observations in mixed-phase clouds.

Data availability

The CAESAR dataset provides open-access observations of cold-air outbreak clouds over the Nordic Seas (EOL archive). Collaboration with Elise Rosky began during a 2025 visit to NSF NCAR, contributing to a joint AGU 2025 abstract on mixed-phase isotopes (Rosky et al. 2025). The dataset serves as a key benchmark for model evaluation.

4 Project Tasks, Milestones and Schedule

The task structure follows the three work packages (A1–A3), ensuring tight coupling between theory, numerics, and observations. Each task produces at least one research output (publication and/or software release).

A1 Theory: Isotopic Cloud Microphysics

- T1** Derive governing equations for isotope-resolved mass transfer in mixed-phase clouds. Include equilibrium and kinetic fractionation and phase-change processes in a particle-based formulation.

Deliverable: theoretical framework + conference presentation (EGU, M6–12).

- T2** Identify limiting regimes of isotope evolution (equilibrium, Rayleigh, diffusion-limited growth) and construct a regime map with reduced-order models. **Deliverables:** reduced-order models + closure breakdown criteria. **Dissemination:** EGU poster/talk (M12–18).
- T3** Quantify sensitivity of isotopic composition using local and global methods and develop an uncertainty propagation framework linking model parameters to isotopic observables. **Deliverables:** ensemble UQ framework + reproducible notebooks. **Dissemination:** ICCP/EGU (M18–24).

A2 Numerical Framework: Particle-Based Implementation

- T4** Implement isotope-resolved microphysics in PySDM for vapour, liquid, and ice phases, ensuring conservation and consistency with analytical limits. **Deliverable:** open-source PySDM isotope module (GitHub + DOI). **Dissemination:** EGU demo (M12–18).
- T5** Develop a minimal isotope-enabled cloud model for controlled experiments isolating key drivers of isotopic closure (supersaturation, mixing, phase partitioning). **Deliverables:** full mixed-phase extension + verified isotope formulation + documented API. **Dissemination:** tutorial/summer school materials (M14–20).

A3 Validation and Closure Assessment

- T6** Prepare observational and laboratory datasets (CAESAR, NCAR), including uncertainty quantification and mapping to model variables. **Deliverable:** curated model-ready dataset + preprocessing pipeline.
- T7** Perform model–data comparison to evaluate isotopic closure and identify microphysical processes required to reproduce observed isotopic signatures. **Deliverables:** closure assessment framework + classification of closure regimes. **Dissemination:** final journal paper + full open-source release (M18–24).

Milestones

- M1** Completion of theoretical framework for isotope-resolved cloud microphysics and identification of governing equations (Month 12).
- M2** Completion of numerical implementation in PySDM and validation of isotope tracking in idealised simulations (Month 18).
- M3** Integration of observational and laboratory datasets with model framework and initiation of systematic evaluation (Month 12).
- M4** Completion of isotopic closure assessment and identification of minimal physical requirements for closure (Month 24).

Schedule

The project is structured over 24 months with strong feedback between theory, numerics, and data analysis. Dataset preparation starts early, while closure analysis is performed in the final phase.

Result dissemination

Project results will be disseminated through peer-reviewed publications, conference presentations, research visits, and open-source software releases. Manuscripts will be submitted to journals in atmospheric modelling and isotope hydrology, including *Tellus B: Chemical and Physical Meteorology*, *Journal of Advances in Modeling Earth Systems*, and *Geoscientific Model Development*. Interim and final results will be presented at international conferences, such as the European Geosciences Union General Assembly and the International Conference on Clouds and Precipitation. The project includes a research visit to the United

States, a visit from an international collaborator to AGH University of Krakow, and participation of the MSc student in an international summer school. All developed software will be released as open source via GitHub, together with documentation and reproducible Jupyter notebooks demonstrating the methods and simulations. Data-processing workflows required to reproduce published results will also be made publicly available.

5 Research methodology

The methodology combines physically based modelling, particle-based simulation, and data-model comparison.

PySDM-based modelling framework

The model is based on PySDM (Super-Droplet Method), extended with prognostic isotopic variables in vapour, liquid, and ice phases. This preserves the Lagrangian particle structure while enabling isotope-resolved microphysics. Simulations are performed in idealised box, parcel, and column setups.

Reproducible simulation environment

Jupyter notebooks provide the main interface for model development and analysis. Each notebook defines a complete workflow (setup, execution, diagnostics, visualisation), ensuring full reproducibility without external scripts.

Notebooks are integrated with version control and CI pipelines that verify physical consistency, numerical stability, and benchmark agreement. This ensures transparent and reproducible model development.

Training component

The workflow supports student training through simplified simulation notebooks, reproducible experiments, and analysis tools, providing hands-on experience in cloud microphysics and scientific computing.

Composition and qualifications of the research team

The team consists of the Principal Investigator (PI), a scientific mentor, and one student.

Principal Investigator The PI is a PhD student in physical sciences with a background in mathematics and numerical modelling. They specialise in scientific computing and particle-based atmospheric simulation frameworks. The PI develops the isotope-enabled cloud microphysics model in this proposal, integrating theory, numerics, and data analysis. Previous work includes isotope-enabled modelling and collaboration at NCAR with Dr. Elise Rosky on mixed-phase cloud processes.

Scientific mentor The mentor, Dr. Sylwester Arabas, is co-developer of PySDM and an expert in Lagrangian cloud modelling and scientific computing. He ensures physical and numerical robustness of the framework and contributes to its development.

Student contribution The student will develop a minimal computational environment for isotope-enabled cloud simulations (Python/Julia), supporting research and training in cloud microphysics.

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